

up the development process. Through open source, new technology becomes available faster than the traditional market would allow, and this helps the entire industry address many of today's complex technical issues together.

There is a passion for sharing that drives a lot of the work our teams here do. It also inspires us to find new ways to build and evolve those teams.

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Q What challenges did you face in finding appropriate talent?

The great thing that we found about open source is that it actually is a huge help when searching for the right talent and in building strong connections with engineers that share our ambitions and like solving the same problems as us. Since the people working on our open source projects are the same engineers that are building our production-grade technology, engineers interested in working at Facebook already have a good sense of how the team operates, and vice versa.

We also conduct a survey when new employees join the company, and one of the questions we ask is whether open source was one of the deciding factors for them coming to work at Facebook. We are always proud when we hear it has been!

Q Is India already a source of talent for various open source developments, or do Indian developers still need to acquire some vital skills to join the team under your leadership?

APAC is one of the fastest-growing regions for developers building with Facebook, outside of the US. So developer skillsets in the area are critical to Facebook. As more people come online every day, we want to help developer communities grow to more regions across the globe. To help with this, we created FbStart, a program designed to help early stage mobile startups build and grow their applications, which now has more than 9,000 members spanning 137 countries. India provides the most FbStart members in APAC.

Q Do you think that open source is efficient enough for a social network like Facebook?

The key to ensuring that our projects are easy to use and work with externally is to make them well-architected, modular, and with clean APIs. This means we must write better software in the first place, and then make our own code-reuse and maintainability much better.

It is also important to continuously improve the documentation for each open source project. Proper documentation helps ensure that those who are unfamiliar with the technology can adopt it quickly. This also helps when new engineers inside the company want to ramp up on a project quickly.

Q How do you choose the open and closed source mix?

This is the question we constantly revisit. It can be easy for companies to turn their open source portfolios into a dumping ground sometimes — meaning that code is uploaded once but not maintained. This is often a result of not having the right tools and program management in place.

At Facebook, we want to grow the open source community and to ensure we are moving forward, we make sure that every project we plan to open source will be well maintained and managed. This is one major criterion, but we also tend to focus on projects that we think offer something new and useful for the industry rather than just reinventing the wheel, for the sake of it.

It is worth pointing out that we also have a shutdown process for projects that are no longer valid or well maintained. This allows us to be clear about their status without stopping the community from taking them over and further developing them.

Q Don't you think that deploying open source comes with some serious security concerns? How would you overcome all this?

Openness and security are not mutually exclusive. In fact, open source generally makes your code more secure. By making the code open to the public, developers can help spot vulnerabilities and bugs in your code that you might have otherwise missed.

Q Facebook has now emerged as the leading artificial intelligence (AI) company. How can open source take concepts like AI and deep learning to new levels?

We are excited to pursue projects in many areas and industries beyond traditional developer platforms, including artificial intelligence and machine learning. The industry is starting to recognise that these areas can help us leverage the power of automation to build more quickly and improve the overall performance of our products. Through the power of artificial intelligence and machine learning, we can think more creatively about how we can continue to accelerate the pace of innovation and build breakthrough technology.

And we are applying many of these concepts into our own software at Facebook. For example, we open sourced Torch implementations and training scripts of deep neural nets for image recognition as well as Torchnet — a modular collection of boilerplate code, key abstractions, and reference implementations that build on top of the Torch framework.

We are eager to see how intelligent tooling can help engineers and developers work more efficiently than they do today, at scale.

Q So can we say that open source is the key to Facebook's success?

There are so many things that contribute to Facebook's success, but we are certainly proud to have been able to build alongside — and play our part in contributing back to — the open source movement. Here is to continuing to do so! 